

THIWANKA LAKSHAN

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PROFILE SUMMARY

Undergraduate in Information and Communication Technology at the University of Vavuniya with a strong foundation in full-stack development, web technologies, and software engineering principles. Skilled in building and testing software solutions using React.js, Vue.js, Node.js, and Express.js, with hands-on experience in IoT-based systems, RESTful APIs, and database-driven applications. Strong problem-solving mindset with a commitment to continuous learning and delivering efficient, user-focused solutions.

EDUCATION

University of Vavuniya

June 2023 – Present

Bachelor of Information and Communication Technology Honours (BICT Hons) Degree

PROJECTS

MooMap – Real-Time Cattle Monitoring using IoT Smart Collars & LoRa

([View](#))

Technologies: Node.js, Express.js, PostgreSQL, REST APIs

- Designed and implemented RESTful APIs using Node.js and Express.js to support real-time GPS data ingestion from IoT smart collars.
- Built and managed PostgreSQL database schemas for efficient storage and retrieval of live tracking data.
- Developed scalable server-side logic for real-time data processing, alerting, and secure delivery to web and mobile clients.

FloodSense – Real-Time Flood Prediction and Early Warning System

([View](#))

Technologies: Node.js, Express.js, React, Vue.js

- Developed a Node.js/Express backend paired with React and Vue.js frontends to deliver real-time flood risk alerts.
- Engineered a 5-factor rule-based algorithm achieving 85% prediction accuracy for short-term forecasts.
- Automated weather data collection every 30 minutes across 20+ locations in Colombo, Gampaha, and Kalutara districts, with automated email alerts giving 2–24 hour evacuation lead time.

TECHNICAL SKILLS

Programming Languages: JavaScript, Python, Java

Frontend: React.js, Vue.js, HTML5, CSS3

Backend: Node.js, Express.js, RESTful API Design

Databases: PostgreSQL, MySQL, MongoDB

Tools & Technologies: Docker, Kubernetes, Git, GitHub, Vercel, Render, Postman

Core Concepts: OOP, Data Structures & Algorithms, SDLC, Agile

SOFT SKILLS

Problem Solving, Technical Project Leadership, Teamwork, Adaptability, Continuous Learning, Analytical Thinking

CERTIFICATIONS

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|--|----------------|--------------------------|
| Developing Back-End Apps with Node.js and Express – IBM | October 2025 | (View) |
| Developing Front-End Apps with React – IBM | December 2025 | (View) |
| Getting Started with Git and GitHub – IBM | September 2025 | (View) |
| Introduction to Containers w/ Docker, Kubernetes & OpenShift – IBM | June 2026 | (View) |
| Python for Data Science, AI & Development – IBM | April 2026 | (View) |

EXPERIENCE & VOLUNTEERING

Vice Chair

IEEE Computer Society, University of Vavuniya ·

October 2025 – Present

- Led as project Vice Chair for IEEE Andaharaya 2025, spearheading a nationwide program to increase IEEE membership awareness and engagement.
- Represented the IEEE Student Branch of the University of Vavuniya at IEEE LeaderConn 2026, gaining leadership and team management experience.

Team Lead

IEEE Student Branch, University of Vavuniya ·

April 2025 – Present

- Directed Program and Logistics for IEEE Day 2025, managing event execution and developing leadership and team coordination skills.
- Captained Team BugBusters9588 at IEEEExtreme 19.0 (24-hour global coding contest), competing against 19,000+ participants and applying problem-solving, teamwork, and decision-making under pressure.

REFERENCES

Mr. V. Senthooran

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